`genQE` — Two-Nucleon GCF Event Generator

Physics & Pipeline Documentation

GCF with 3N Project

October 21, 2026

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1 Overview

`genQE` is a Monte Carlo event generator for quasi-elastic (QE) electron scattering off nuclei within the **Generalized Contact Formalism (GCF)**. It generates events of the type

$$e + A \longrightarrow e' + N_{\text{lead}} + N_{\text{rec}} + (A{-}2)^*, \tag{1}$$

where the electron scatters off one nucleon of a short-range correlated (SRC) pair. The struck nucleon (N_{lead}) carries the virtual-photon momentum; the partner (N_{rec}) recoils, and the residual ($A{-}2$) system is left with excitation energy E^*.

The generator factorizes the problem into two independent parts:

1. **Nuclear structure:** sampling the SRC pair's internal momenta via the GCF (contact coefficients $\times$ universal functions).
2. **Scattering dynamics:** computing the electron–nucleon cross section at the sampled kinematics.

2 Generalized Contact Formalism

2.1 Contact factorization

The GCF expresses the two-body momentum distribution of a nucleus A as

$$n_A^{(ij)}(\boldsymbol{k}_{\text{rel}}) \;=\; C_{ij}^A \left|\varphi_{ij}(k_{\text{rel}})\right|^2, \tag{2}$$

where i, j label isospin–spin channels, C_{ij}^A are the *nuclear contact coefficients* (nucleus-dependent, extracted from *ab initio* calculations or data), and $\varphi_{ij}(k)$ are *universal* (nucleus-independent) functions determined by the short-range behavior of the chosen NN potential.

Four channels are implemented:

Channel	Pair	Quantum numbers
`pp0`	proton–proton	$^1S_0$
`nn0`	neutron–neutron	$^1S_0$
`pn0`	proton–neutron	$^1S_0$
`pn1`	proton–neutron	$^3S_1$ (deuteron-like, dominant)

For identical-nucleon pairs (pp or nn) the spectral function used is

$$S(k) = 2\,C\;\varphi^2(k), \tag{3}$$

while for pn pairs it is

$$S(k) = C_{pn0}\,\varphi_{pn0}^2(k) + C_{pn1}\,\varphi_{pn1}^2(k). \tag{4}$$

The universal functions $\varphi^2(k)$ are tabulated in 100 bins covering $k \in [0,\, 10]\,\text{fm}^{-1}$ with $\Delta k = 0.1\,\text{fm}^{-1}$ and linearly interpolated at runtime.

2.2 Available NN interactions

Several sets of $\varphi^2(k)$ are included, but an interaction can only be run for nuclei that also have contact values for it:

- **AV18** — Argonne v_{18} (default)
- **N2LO_10** — Chiral N^{2}LO, $R = 1.0\,\text{fm}$
- **N2LO_12** — Chiral N^{2}LO, $R = 1.2\,\text{fm}$

- **N3LO_600** — Chiral N^{3}LO, $\Lambda = 600\,\mathrm{MeV}$; universal functions only. No nucleus has contacts for it, so selecting it aborts (exit code -2).
- **AV4Pc** — Argonne AV4′
- **NV2_1a** — Norfolk NV2+Ia; contacts only for ^{2}H, ^{3}He and ^{4}He
- **AV18_deut** — AV18 with deuteron-specific wave function (no contact dependence)

2.3 Contact coefficients

Contact values C_{ij}^A are stored per nucleus and NN interaction. They are loaded on initialization by `gcfSRC::set_Contacts()`, which takes the first set that exists for the nucleus and interaction, in the order: k-space values from the Scale-and-Scheme paper, r-space values from the same paper, the deuteron distribution (`AV18_deut`, no contacts), and finally contact ratios from fits to EG2 data. For AV18 this gives k-space contacts for ^{2}H through ^{12}C, r-space contacts for ^{16}O, ^{40}Ca and ^{40}Ar, and the EG2 fallback for ^{27}Al, ^{56}Fe and ^{208}Pb (see issue #64). The values used for AV18, ^{12}C (k-space) are

$$C_{pp0} = C_{nn0} \approx 6.84, \qquad C_{pn0} \approx 7.46, \qquad C_{pn1} \approx 95.3; \tag{5}$$

for comparison, the r-space set is $C_{pp0} = C_{nn0} = C_{pn0} \approx 4.30$, $C_{pn1} \approx 78.8$, and the EG2 set is $C_{pp0} = C_{nn0} = C_{pn0} \approx 1.07$, $C_{pn1} = 16$. The $pn1$ (deuteron-like) channel dominates by roughly an order of magnitude.

3 Nuclear Parameters

Each target nucleus is described by a `gcfNucleus` object that stores:

- A, Z, nuclear mass m_A
- σ_{CM}: width (GeV/c) of the Gaussian center-of-mass momentum distribution of the pair
- E^*_{max}: maximum excitation energy of the $(A-2)$ residual
- $(A-2)$ masses: m^{pp}_{A-2}, m^{pn}_{A-2}, m^{nn}_{A-2} (depend on which pair is removed)

Nucleus	A	σ_{CM} (GeV/c)	E^*_{max} (GeV)	Note
^{2}H	2	0	0	No CM motion, no residual
^{3}He	3	0.10	0	$(A-2)$ = single nucleon
^{4}He	4	0.10	0.010	$(A-2)$ = deuteron (pn) or $2N$ (pp/nn)
^{12}C	12	0.15	0.030	
^{56}Fe	56	0.15	0.050	
^{208}Pb	208	0.15	0.100	

The excitation energy can be sampled from a Gaussian $E^* \sim \mathcal{N}(E^*_{\mathrm{mean}}, \sigma_E)$ or held fixed.

4 Event Generation Pipeline

4.1 Step 1: Pair type selection

The lead and recoil isospins are drawn independently, each p or n with probability 1/2, so the pair is pp or nn with probability 1/4 each and pn (either ordering) with probability 1/2. The event weight carries an overall factor of 4 (2 nucleons $\times$ 2 possible isospin assignments).

4.2 Step 2: Decay function — sampling the SRC pair

The SRC pair's kinematics are generated in `gcfGenerator::decay_function()`.

Center-of-mass momentum. Each Cartesian component of $\boldsymbol{P}_{\mathrm{CM}}$ is sampled independently from a Gaussian:

$$P_{\mathrm{CM},\alpha} \sim \mathcal{N}(0,\ \sigma_{\mathrm{CM}}). \tag{6}$$

Relative momentum. The relative momentum $\boldsymbol{k}_{\mathrm{rel}}$ is sampled uniformly:

$$k \in [k_{\min},\, k_{\max}], \quad \cos\theta_{\mathrm{rel}} \in [-1,\, 1], \quad \phi_{\mathrm{rel}} \in [0,\, 2\pi]. \tag{7}$$

Default range: $k_{\min} = 0.2\,\mathrm{GeV}/c$, $k_{\max} = 1.05\,\mathrm{GeV}/c$.

Weight from spectral function. The phase-space weight is

$$w_{\mathrm{decay}} = \Delta k\ \Delta(\cos\theta)\ \Delta\phi\ \times\ \frac{k^2\, S(k,\, i, j)}{(2\pi)^3}, \tag{8}$$

where $S(k)$ is the GCF spectral function from Eq. (2).

Pauli blocking cut. If $k < k_{\mathrm{cut}}$ (default $0.25\,\mathrm{GeV}/c$; 0 for deuterium), the event is rejected ($w = 0$).

Individual momenta.

$$\boldsymbol{p}_{\mathrm{lead}} = \tfrac{1}{2}\boldsymbol{P}_{\mathrm{CM}} + \boldsymbol{k}_{\mathrm{rel}}, \qquad \boldsymbol{p}_{\mathrm{rec}} = \tfrac{1}{2}\boldsymbol{P}_{\mathrm{CM}} - \boldsymbol{k}_{\mathrm{rel}}. \tag{9}$$

Lead nucleon energy. From energy conservation:

$$E_{\mathrm{lead}} = m_A - E_{A-2} - E_{\mathrm{rec}}, \qquad E_{\mathrm{rec}} = \sqrt{|\boldsymbol{p}_{\mathrm{rec}}|^2 + m_N^2}, \tag{10}$$

where $E_{A-2} = \sqrt{|\boldsymbol{p}_{A-2}|^2 + m_{A-2}^2}$ and $\boldsymbol{p}_{A-2} = -\boldsymbol{P}_{\mathrm{CM}}$.

4.3 Step 3: Light-cone variant

An alternative light-cone kinematic mode (`decay_function_lc`) is available. It uses light-cone momentum fractions

$$\alpha = \frac{p^+}{\bar{m}}, \qquad \bar{m} = m_A/A, \tag{11}$$

with transverse momentum $\boldsymbol{p}_\perp$. The relative light-cone momentum fraction α_{rel} is sampled uniformly, and the constraint

$$k_{\min} = \frac{|1 - \alpha_{\mathrm{rel}}|}{\sqrt{\alpha_{\mathrm{rel}}(2 - \alpha_{\mathrm{rel}})}}\, m_N \tag{12}$$

is enforced to keep kinematics physical. The weight is

$$w_{\mathrm{lc}} = \Delta\alpha\ \Delta k\ \Delta\phi\ \times\ \frac{k\sqrt{k^2 + m_N^2}\ S(k)}{(2\pi)^3}. \tag{13}$$

4.4 Step 4: Electron scattering vertex

Given the lead nucleon's 4-momentum from Step 2, the scattered electron kinematics are determined.

Sampling. Two variables are drawn uniformly:

$$Q^2 \in [Q^2_{\min},\, Q^2_{\max}], \qquad \phi_k \in [0,\, 2\pi]. \tag{14}$$

Kinematic constraint. The electron's transverse momentum $k_\perp$ is determined by solving a quadratic arising from energy–momentum conservation of the full reaction. Let $p_1^- = E_1 - p_{1z}$ denote the nucleon's light-cone minus component and $k^- = Q^2/(2E_{\rm beam})$. Then

$$A_e = \frac{k^-}{p_1^-}, \qquad c_e = A_e\big(p_1^+\, k^- - 2E_{\rm beam}\, p^-_{\rm lead} - \mathrm{virt}\big), \qquad b_e = -2A_e\, y, \tag{15}$$

where $y = p_{1\perp}\cos(\phi_k - \phi_1)$ and "virt" encodes off-shell effects. The solution is

$$k_\perp = \frac{-b_e \pm \sqrt{b_e^2 - 4c_e}}{2}. \tag{16}$$

Events with $D = b_e^2 - 4c_e < 0$ are rejected. When two physical solutions exist, both are equally probable and the weight is multiplied by 2.

Jacobian. The δ-function constraint contributes a Jacobian factor

$$J = 2\,E_{\rm beam}\,E_k\left|1 - \frac{p_{1z} + y\tan(\theta_k/2) + E_{\rm beam} - E_k}{E_{\rm lead}}\right|, \qquad w \;*=\; 1/J. \tag{17}$$

4.5 Step 5: Cross section

The electron–nucleon cross section σ_{eN} is evaluated at the sampled kinematics. Three prescriptions are available:

On-shell. The Rosenbluth cross section for a free nucleon at rest,

$$\sigma_{\rm Ros} = \frac{\alpha^2}{4E^2_{\rm beam}\sin^4(\theta/2)}\,\frac{E'}{E_{\rm beam}}\left[\frac{G_E^2 + \tau G_M^2}{1+\tau}\cos^2\frac{\theta}{2} + 2\tau G_M^2\sin^2\frac{\theta}{2}\right], \tag{18}$$

with $\tau = Q^2/(4m_N^2)$, is returned in de Forest's σ_{eN} normalisation, which carries no recoil or flux factor (his Eq. (13)):

$$\sigma_{\rm onshell} = \sigma_{\rm Ros}\,\frac{E_{\rm beam}}{E'\,(1+2\tau)}. \tag{19}$$

This equals CC1 for a nucleon at rest, and is about $0.47\,\sigma_{\rm Ros}$ at 6 GeV, 25°. Only the electron angle θ is taken from the event: $E' = E_{\rm beam}m_N/[m_N + E_{\rm beam}(1-\cos\theta)]$ and $Q^2 = 2E_{\rm beam}E'(1-\cos\theta)$ are recomputed from elastic kinematics, so τ and the form factors are evaluated at the elastic Q^2, not the event's (e.g. 4.2 instead of $4.7\,\mathrm{GeV}^2$ at $x_B = 1.38$).

CC1 and CC2 (de Forest). Both are de Forest's off-shell extrapolations (Nucl. Phys. A 392 (1983) 232). They use the same current conservation, $\boldsymbol{q}\cdot\boldsymbol{J} = \omega\rho$ (his Eq. (9)), which puts the longitudinal current in the Coulomb gauge, and the same cross section (his Eq. (13)),

$$\sigma_{eN} = \sigma_{\rm Mott}\Big[\tfrac{Q^4}{q^4}\,w_C + \big(\tfrac{Q^2}{2q^2} + \tan^2\tfrac{\theta}{2}\big)w_T + \tfrac{Q^2}{q^2}\big(\tfrac{Q^2}{q^2} + \tan^2\tfrac{\theta}{2}\big)^{1/2}w_I\cos\phi + \big(\tfrac{Q^2}{q^2}\cos^2\phi + \tan^2\tfrac{\theta}{2}\big)w_S\Big], \tag{20}$$

with $\sigma_{\rm Mott} = 4\alpha^2E'^2\cos^2(\theta/2)/Q^4$, $q = |\boldsymbol{q}|$, and ϕ the angle between the scattering plane and the $(\boldsymbol{p}_{\rm lead}, \boldsymbol{q})$ plane. They differ only in the off-shell nucleon vertex from which $w_{C,T,S,I}$ are built (using F_1 and κF_2 obtained from G_E, G_M):

- **CC1** (default): the Gordon-reduced vertex $\gamma_\mu(F_1+\kappa F_2)+i(p'+p)_\mu\kappa F_2/2m_N$ (his Eq. (16)), giving the structure functions of his Eq. (17). These are the free ones with $\bar{E}=\sqrt{p^2+m_N^2}$ ($\boldsymbol{p}=\boldsymbol{p}_{\rm lead}-\boldsymbol{q}$) and $\bar{\omega}=E_{\rm lead}-\bar{E}$ in place of E and ω.
- **CC2**: the vertex $\gamma_\mu F_1-\sigma_{\mu\nu}q_\nu\,\kappa F_2/2m_N$ (his Eq. (15)), giving the structure functions of his Eq. (18).

4.6 Step 6: Form factors

Three electromagnetic form-factor models are implemented in `eNCrossSection`, but every driver (`genQE`, `genQE_EIC`, …) hard-codes `kelly`, so dipole and Ye cannot be selected without editing the source.

Dipole. $G_E^p=G_D=(1+Q^2/0.71)^{-2}$, $G_M^p=\mu_p\,G_D$.

Kelly (used). Rational parameterization:

$$G(Q^2)=\frac{1+a_1\tau}{1+b_1\tau+b_2\tau^2+b_3\tau^3}, \tag{21}$$

with fit parameters given separately for G_E^p, G_M^p/μ_p and G_M^n/μ_n. G_E^n uses Kelly's Galster form instead,

$$G_E^n=\frac{A\tau}{1+B\tau}\,G_D, \qquad A=1.70, \quad B=3.30. \tag{22}$$

Ye. 13-parameter z-expansion fit.

4.7 Step 7: Final weight

The total event weight is

$$w=4\;\times\;w_{\rm decay}\;\times\;\left(\Delta Q^2\cdot 2\pi\right)\;\times\;\frac{1}{J}\;\times\;\sigma_{eN}. \tag{23}$$

4.8 Step 8: Optional corrections

Coulomb correction. The electron energy is shifted before and after scattering by the nuclear Coulomb potential, taken as 0.775 of the potential at the centre of a uniformly charged sphere:

$$\Delta E_{\rm Coulomb}=0.775\times\frac{3}{2}\,\frac{Z\alpha_{\rm em}\,\hbar c}{R_C}, \qquad R_C=\left(1.1\,A^{1/3}+0.86\,A^{-1/3}\right)\text{fm}, \tag{24}$$

and $\Delta E_{\rm Coulomb}=0$ for ^{2}H. The beam gains $\Delta E_{\rm Coulomb}$ before the vertex; afterwards the scattered electron loses it and outgoing protons gain it.

Bremsstrahlung. Radiative corrections via the peaking approximation. The incoming and the scattered electron each lose a single energy ΔE (one emission, not a Poisson number of photons), with $\Delta E/E=r^{1/\lambda}$ for uniform $r\in[0,1]$, i.e.

$$P(\Delta E/E)\propto\left(\frac{\Delta E}{E}\right)^{\lambda-1}, \qquad \lambda=\frac{\alpha}{\pi}\left(\ln\frac{4E^2}{m_e^2}-1\right). \tag{25}$$

The weight is multiplied by $(1-\delta_{\rm hard})\left(E_{\rm beam}/\sqrt{E_{\rm beam}E'}\right)^{\lambda_i}\left(E'/\sqrt{E_{\rm beam}E'}\right)^{\lambda_f}$, with $\lambda_{i,f}$ evaluated at $E_{\rm beam}$ and E' and $\delta_{\rm hard}=\frac{2\alpha}{\pi}\left(-\frac{13}{12}\ln\frac{Q^2}{m_e^2}+\frac{8}{3}\right)$.

5 Output

Events are stored in a ROOT `TTree` named `genT` with branches:

Branch	Type	Description
`lead_type`	int	PDG code (2212 = p, 2112 = n)
`rec_type`	int	PDG code of recoil nucleon
`pe[3]`	double	Scattered electron $\boldsymbol{p}_{e'}$
`pLead[3]`	double	Lead nucleon $\boldsymbol{p}_{\text{lead}}$
`pRec[3]`	double	Recoil nucleon $\boldsymbol{p}_{\text{rec}}$
`pAm2[3]`	double	$(A-2)$ momentum
`pRel[3]`	double	Relative momentum $\boldsymbol{k}_{\text{rel}}$
`q[3]`	double	Momentum transfer $\boldsymbol{q}$
`weight`	double	Event weight

6 Summary of Algorithm

For each Monte Carlo trial:

1. Draw lead and recoil isospins independently (pp, nn: 1/4 each; pn: 1/2).
2. Sample $\boldsymbol{P}_{\text{CM}}$ (Gaussian) and $\boldsymbol{k}_{\text{rel}}$ (uniform in k, $\cos\theta$, ϕ).
3. Evaluate GCF spectral function $S(k)$; apply Pauli cut.
4. Derive individual nucleon momenta and $(A-2)$ recoil.
5. (Optional) Apply Coulomb shift and bremsstrahlung to beam.
6. Sample Q^2 and ϕ_k; solve quadratic for $k_\perp$.
7. Compute Jacobian J from energy conservation constraint.
8. Evaluate σ_{eN} (CC1/CC2/on-shell, with Kelly form factors).
9. Multiply all weight factors; store event if $w > 0$.